

Period 3 Chlorides & Oxides

(Cambridge (CIE) A Level Chemistry):

Revision Note

Bonding in Period 3 Chlorides & Oxides

Period 3 chlorides

Chemical bonding & structure of Period 3 chlorides table

Period 3 chloride	NaCl	MgCl ₂	Al ₂ Cl ₆	SiCl ₄	PCl ₅	SCl ₂
Chemical bonding	Ionic	Ionic	Covalent	Covalent	Covalent	Covalent
Structure	Giant ionic	Giant ionic	Simple molecular	Simple molecular	Simple molecular	Simple molecular

Period 3 oxides

Chemical bonding & structure of Period 3 oxides table

Period 3 oxide	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀	SO ₂ SO ₃
Chemical bonding	Ionic	Ionic	Ionic (with some covalent character)	Covalent	Covalent	Covalent
Structure	Giant ionic	Giant ionic	Giant ionic	Giant covalent	Simple molecular	Simple molecular

- Going across Period 3, their chlorides and oxides become more **covalent** and their structure shifts from a **giant ionic** to a **simple molecular** structure
- Their reactions with water become more **vigorous** as a result of this, as it becomes easier to **hydrolyse** the chlorides and oxides